

So, if you start 10 years late, you need to invest almost 3 times more per year and if you start 30 years late, do you think you need to invest 9 times more? No. It is a huge 44 times more per annum to reach to your target.

When it comes to the effects of compounding, you can never make up for the lost time.

Let us look at the example shown in Fig 5.2. Interest starts overtaking the principle after some time and becomes multiple fold very soon.

If you remember the Formula 3 that explains CAGR from our earlier chapter, you would appreciate that "Time" is the most crucial factor in the entire equation. Compounding is extraordinary and time and discipline are the main ingredients needed for this extraordinary impact. The table shows an investment of ₹6,000 earning a cumulative interest of 8% per year.

Year	Principle	Interest @ 8%	Accumulated Principle + Interest
1	6,000	480	6,480
2	6,480	518	6,998
3	6,998	560	7,558
4	7,558	605	8,163
5	8,163	653	8,816
6	8,816	705	9,521
7	9,521	762	10,283
8	10,283	823	11,106
9	11,106	888	11,994
10	11,994	960	12,954
11	12,954	1,036	13,990
12	13,990	1,119	15,109
13	15,109	1,209	16,318
14	16,318	1,305	17,623
15	17,623	1,410	19,033
16	19,033	1,523	20,556
17	20,556	1,644	22,200
18	22,200	1,776	23,976
19	23,976	1,918	25,894
20	25,894	2,072	27,966
21	27,966	2,237	30,203

Year 10
 Accumulated money becomes 2 times the principle

Year 15
 Accumulated money becomes 3 times the principle

Year 19
 Accumulated money becomes 4 times the principle

Year 21
 Accumulated money becomes 5 times the principle

You can see in Fig 5.2 above that we start with a principle corpus of ₹ 6,000/- in Year 1 and we add another corpus of ₹6,000/- by interest in the Year 11. It took 11 years to add the first ₹6,000/- but the magic has just started. The next ₹6,000 get added in the next 5 years (by Year 15), the next ₹6,000 in next 4 years (Year 19) and the next ₹6,000/- in next 2 years...this is time and compounding working together. The more time you give for compounding to work, the more magical it becomes.